NLP Final Project Report of Part B- News Article Classification.

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BATCH: Data Science (May 15th)

NOTE: I was having a problem my loom ID, So I recorded the video on my friend's ID. Please don't get confused. I am introducing myself in the video.

Project Explanation Video: https://www.loom.com/share/a9d95c6d0d2d4e74a2e3bbfab67661d4

Introduction

Part B focuses on building a multi-class text classification model using a labeled news dataset. The goal is to automatically categorize news articles into topics such as **sports, politics, technology**, etc., using NLP preprocessing and machine learning.

Dataset Description

The dataset contained the following columns:

- **category** – the label of the news article
- **headline** – the title of the news
- **short_description** – short summary of the news
- **links** – source link
- **keywords** – related keywords

For classification, the **short_description** column was used as the primary text input.

Data Preprocessing

The dataset was cleaned and prepared using the following steps:

- Removal of missing values
- Removal of duplicate text entries
- Text cleaning (lowercasing, removing HTML, URLs, non-alphabet characters)
- Tokenization
- Stopword removal
- Stemming using **Porter Stemmer**
- Creation of `clean_text` and `final_text` columns for modeling

This ensured that all text was standardized and ready for feature extraction.

Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset:

- Category distribution bar plot
- Basic inspection of text length and number of samples per class

The dataset showed a balanced distribution across the major categories.

Feature Extraction

TF-IDF was used to convert text into numerical vectors:

- Unigrams and bigrams (`ngram_range=(1,2)`)
- Maximum features set to 5000
- Resulting in a sparse matrix suitable for machine learning models

TF-IDF was chosen due to its strong performance with short news texts.

Model Development & Training

Three machine learning models were developed:

- **Logistic Regression**
- **Multinomial Naive Bayes**
- **Support Vector Machine (Linear SVM)**

Steps followed:

- Train–test split (80:20)
- Training each model on TF-IDF vectors
- Applying **5-fold cross-validation** to ensure robustness

Model Evaluation

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

- Cross-validation scores

Results showed that **SVM** and **Logistic Regression** performed the best, while **Naive Bayes** performed slightly lower due to its simplification assumptions.

Model Comparison

A comparison table was created containing all evaluation metrics.
Visualizations included:

- **Accuracy comparison bar chart**
- **Confusion matrix for each model**

The comparison showed that **Logistic Regression** produced the best overall performance.

Conclusion

Part B successfully demonstrated the use of NLP and machine learning for classifying news articles. TF-IDF combined with Logistic Regression yielded the highest accuracy, making it the best model for this task. The experiment also highlighted the effectiveness of preprocessing and feature extraction in improving model performance.